
Endocrine Pharmacology III – Pharmacology of Weight Management

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Learning goal

To facilitate an understanding of the current and accepted weight loss therapies as well as become familiar with those agents that affect body weight

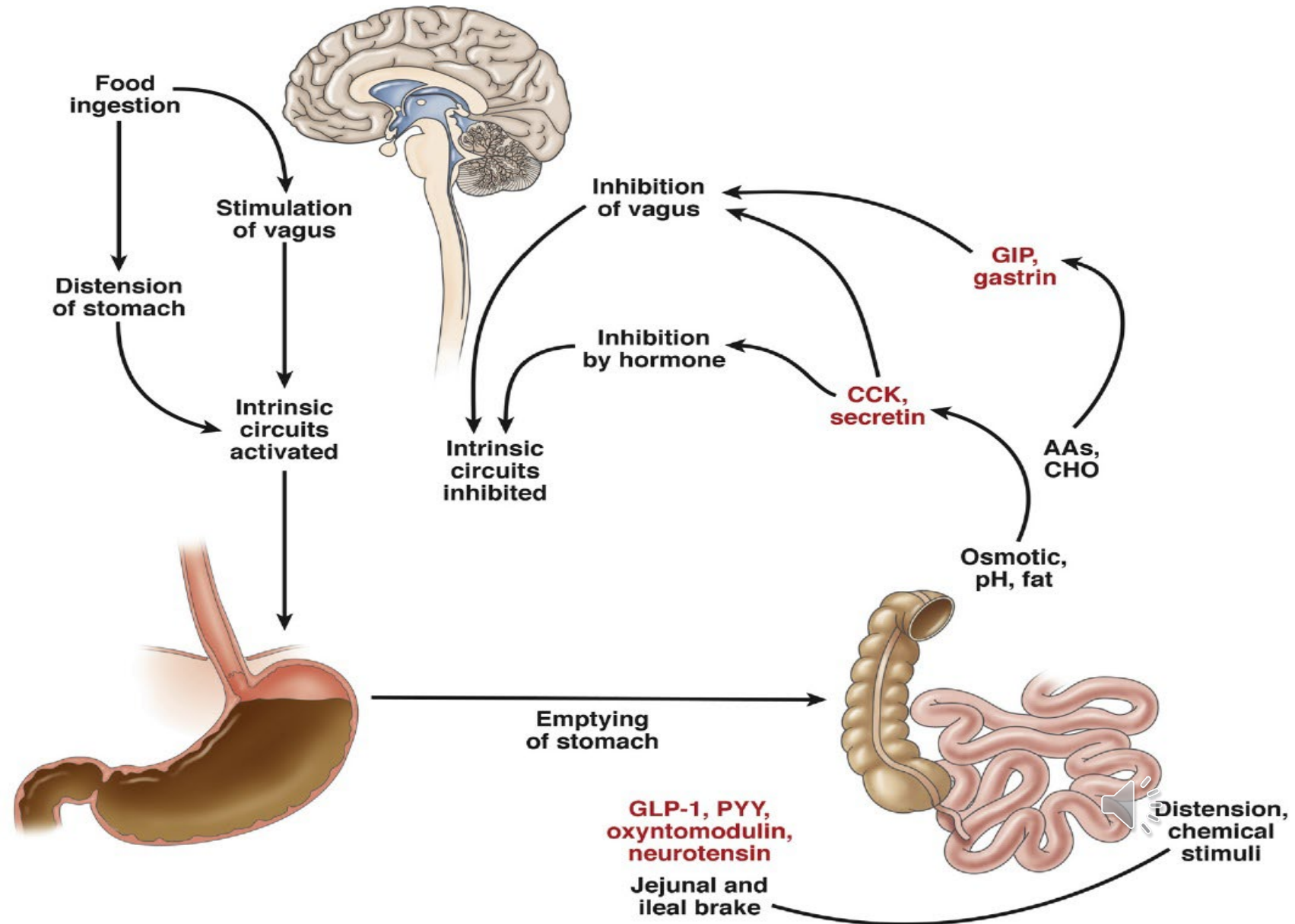


Learning objectives

1. Examine the Fen-Phen and rimonabant controversies through the lens of the FDA review/approval process.
2. Compare and contrast FDA approved drugs for weight loss, including mechanisms of action, pharmacokinetics, and adverse effects.
3. Give examples of drugs that are used to stimulate appetite.



Integration of signals (REVIEW)





Lifestyle interventions



Behavioral therapy



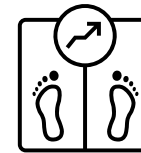
Dietary interventions



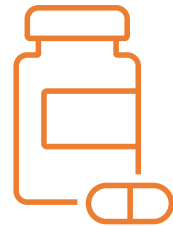
Bariatric procedures



Physical activity



Weight regain



Pharmacotherapy

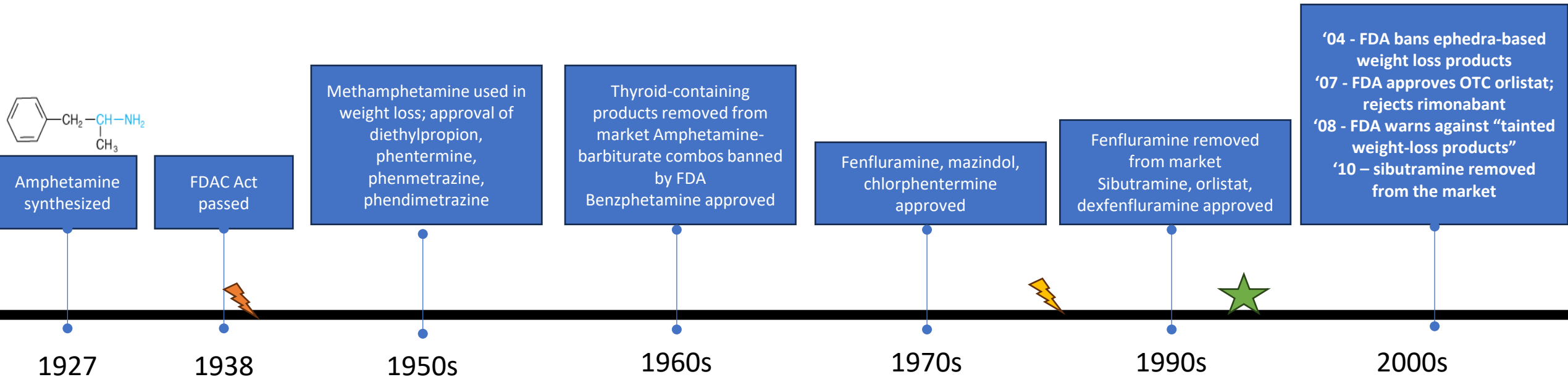


Weight Loss Medications Can Work...

- as a sympathomimetic
- as an anorectic
- to reduce appetite and food/calorie intake
- to prevent fat absorption
- to reduce hepatic gluconeogenesis and insulin production



Did you know?



 Dinitrophenol-containing weight loss products removed from the market

 Phenylpropanolamine removed from market

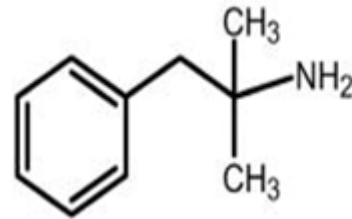
 Leptin discovered



Fen-Phen Controversy

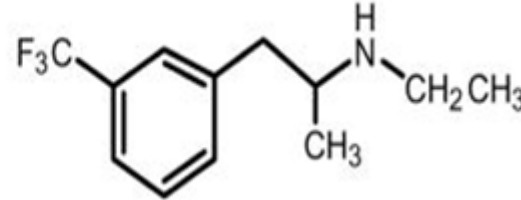
Phentermine

- Promotes DA and NE release
- Inhibits DA and NE reuptake



Fenfluramine

- Promotes 5-HT release
- Inhibits 5-HT reuptake

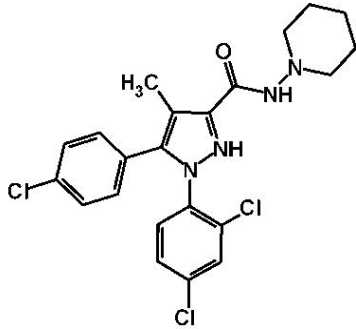


Never combined into one product, but were taken concomitantly

Aortic regurgitation occurred in up to 25% of those taking fenfluramine



Rimonabant Controversy



Cannabinoid-1 (CB-1) receptor antagonist
(*inverse agonist*)
decreases high fat and high sweet food intake

- 2006: Rimonabant approved as a weight-loss medication in Europe and Mexico
- 2007: FDA voted unanimously against approval due to **psychiatric disturbances**
- 2008: Pulled off European market due to same adverse effects

depression
anxiety
changes in sleep patterns
suicidal thoughts

Reported to have effects on insulin sensitivity and endothelial cell function



Sympathomimetic drugs for weight loss

- Work in the CNS where they increase NE in the synaptic cleft and directly stimulate beta-adrenergic receptors.
- Also reduce appetite
- May also increase thermogenesis

Drug	Weight Loss	Common adverse effects
Phentermine	~3.6% - 5.1%	Palpitations, elevated blood pressure
Phentermine/topiramate	~6.6% - 8.6%	Depression, suicidal ideation, CV events, memory loss, birth defects
Diethylpropion	~3.6% - 8%	Palpitations, elevated blood pressure, restlessness, CNS stimulation

Schedule IV drugs

Blood pressure and heart rate should be monitored

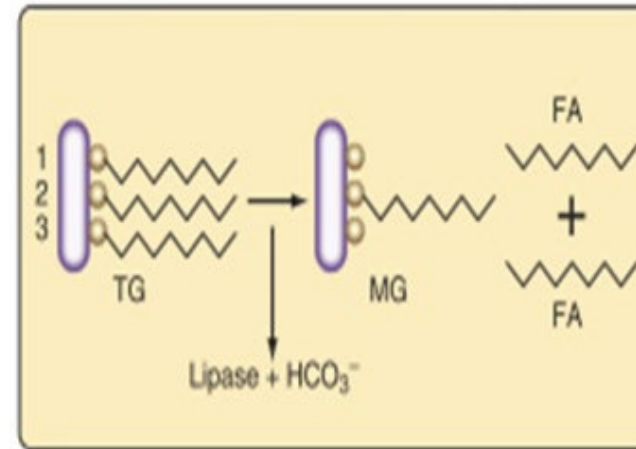
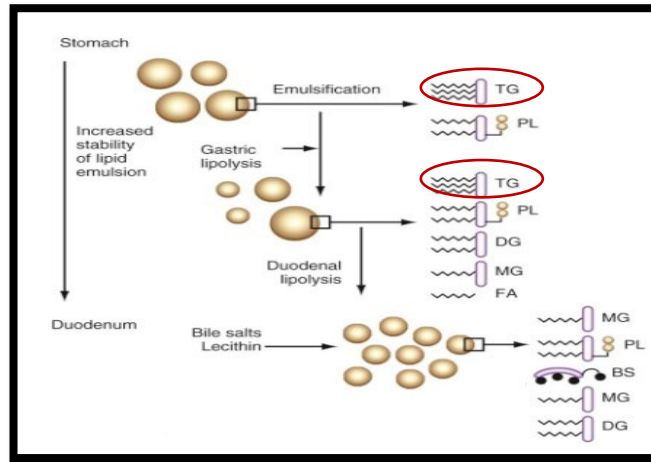
Discontinuation rates due to adverse effects

Abrupt discontinuation should be avoided to prevent withdrawal symptoms after prolonged use



Prevention of fat absorption

- Inhibits pancreatic lipase



Drug	Weight Loss	Common adverse effects
Orlistat	~2.8% - 4%	Fecal incontinence, steatorrhea, diarrhea, flatulence

Not a preferred option due to adverse effects
May carry small risk of cholelithiasis
May reduce absorption of fat-soluble vitamins and certain drugs



Reduction of appetite and cravings

Drug	Weight Loss	Common adverse effects
Naltrexone/bupropion	~3.2%	N/V, vomiting, constipation, diarrhea, headache, dizziness, insomnia, dry mouth

Blood pressure, heart rate, suicidal thoughts/behavior should be monitored

Discontinuation rates due to adverse effects

Avoid in patients taking opioids

Requires dose reduction in the presence of CYP2B6 inhibitors or kidney or liver impairment



GLP-1 and GIP/GLP-1 Receptor Agonists for weight loss

Characteristic	Semaglutide	Liraglutide	Tirzepatide
Description	modified in to provide stabilization against degradation by DPP-4, and to ensure the attachment of only one fatty di-acid	Modified GLP-1 with C-16 palmitoyl fatty acid moiety	a GIP receptor and GLP-1 receptor agonist
Subcutaneous administration	1X weekly	1X daily	1X weekly
Half-life	1 wk	13 h	5 – 6 d
Time to [peak]	1 – 3 d	8-12 h	8 – 72 h
Weight loss	10.3% - 14.4%*	2.7% - 4%	14% - 20%*

Nausea, vomiting, diarrhea, constipation, dyspepsia,
headache, fatigue, dizziness, abdominal pain,
hypoglycemia, acute pancreatitis, acute cholecystitis,
and increased heart rate



*in patients without diabetes

Boxed Warning for GLP-1 Receptor agonists

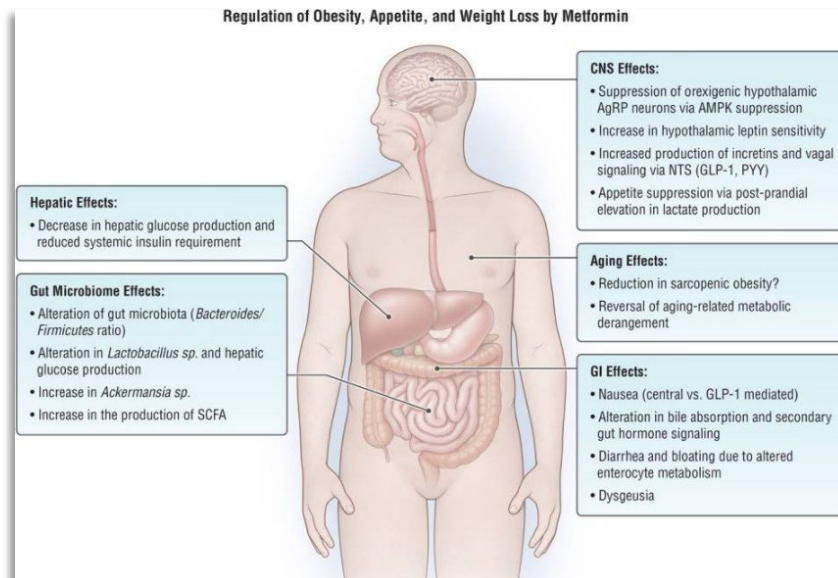
WARNING: RISK OF THYROID C-CELL TUMORS; contraindicated in patients with a personal or family history of MTC or in patients with Multiple Endocrine Neoplasia syndrome type 2 (MEN 2)



Other drugs

Metformin

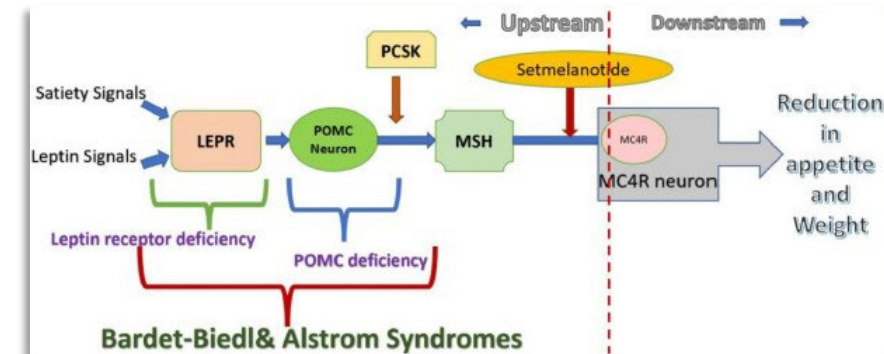
- Works to reduce hepatic gluconeogenesis, insulin production and appetite



Yerevanian and Soukas 2020

Setmelanotide

- A melanocortin 4 receptor agonist
- Approved for the treatment of obesity in patients with hypothalamic dysfunction



Qamar et al. 2024

Weight Gain/Unintentional Weight Loss

- There is a small number of medications that have been used to stimulate appetite, but adverse effects limit their use, moreover, there are no data on the long-term efficacy for improving quality of life or reducing morbidity/mortality.
 - **Dronabinol**, an orally active cannabinoid, is FDA approved for the stimulation of appetite in patients with HIV
 - **Megesterol**, a derivative of progesterone
 - **Cyproheptadine**, an antihistamine



Key points/summary

- The regulation of appetite and satiety is complex and involves the integration of many signals.
- The history of weight loss medications is checkered with product rejections and withdrawals.
- The seven anti-obesity medications approved by the FDA are phentermine, orlistat, phentermine/topiramate ER, naltrexone/bupropion SR, liraglutide, semaglutide and tirzepatide.
- Adverse effects of anti-obesity medications lead to discontinuation.
- GLP-1-induced satiety is via intestinal innervation; dipeptidyl peptidase-4 (DPP-4) inactivates GLP-1
 - GLP-1 receptor agonists can often cause rapid weight loss

